DAI-RONG WU

Munich, Germany · **U.S. Green Card Holder - No visa sponsorship required**

+49-15235852368 · dairong.wu.official@gmail.com · linkedin.com/in/dai-rong-wu · github.com/dairong-wu

CAD Automation | EDA Infrastructure | PDK Development | SPICE Model Enablement

PROFESSIONAL SUMMARY

CAD automation engineer with 8+ years delivering production-scale automation platforms, and process design kits (PDKs) at Tier-1 semiconductor and EDA companies. Reduced engineering cycle times by 50–93% through scalable SKILL/Python automation, cross-foundry PDK delivery, and model full release kit frameworks. Experienced leading cross-functional initiatives with U.S.-based teams across EMEA and APAC. **U.S. Green Card holder and available to relocate immediately, no visa sponsorship required.**

TECHNICAL SKILLS

Automation & Infrastructure: Python (OOP, GUI toolkits), Cadence SKILL, Tcl, Perl, Shell scripting

EDA Platform: Cadence Virtuoso (ADE, Schematic, Layout), Siemens Tanner EDA, Mentor Calibre (DRC/LVS/PEX)

PDK/iPDK Architecture: PCell development, CDF/Callbacks, Netlisting, DRC/LVS deck generation, cross-platform enablement

Simulation Infrastructure: SPICE compact modeling (BSIM4, BSIM-Bulk, HiSIM-HV), HSPICE/Spectre/Eldo, model QA pipelines

DevOps & Workflow: Git, SVN, Jira, Confluence

Semiconductor Domain: CMOS, FinFET, High-Voltage devices, analog/mixed-signal design enablement

PROFESSIONAL EXPERIENCE

Staff Engineer — CAD Automation & Compact Modeling | Infineon Technologies, Munich, Germany *Sep 2023 – Present*

- **Architected and deployed a production model lifecycle platform** consolidating model translation, QA validation, and release pipelines into a unified framework, cutting QA cycle time by 88% (24h → 3h) and eliminating manual error pathways.
- **Developed a scalable gradient-descent parameter centering engine**, reducing manual SPICE tuning time by 93% and accelerating compact model delivery for high-voltage device families.
- **Built and maintained a Virtuoso-integrated validation infrastructure** for automated device characterization, seamlessly handling testbench generation, multi-parameter sweeps, and cross-simulator verification (HSPICE, Spectre, Titan).
- **Directed end-to-end model enablement across two production technology nodes (including SPT9U)**, coordinating foundry partners, PDK, and design teams across EMEA to support global compact-model releases and downstream customer adoption.
- Standardized cross-team model delivery workflows and mentored new hires on modeling infrastructure; **drove engineering execution and resource allocation** to ensure on-time releases across technology nodes.

Software Engineer — EDA & PDK Platform Development | Siemens EDA, Hsinchu, Taiwan *Mar 2020 – Aug 2023*

- **Designed and shipped production PDKs for 5+ foundries across 10+ process technology nodes,** enabling Tanner EDA design flows for global customer design teams.
- **TSMC iPDK Migration & Automation Tooling**: Built a compatibility layer for TSMC iPDKs by migrating legacy SKILL code to Tcl/Python for Tanner tool support. Automated the conversion of symbols and PCells, reducing the total PDK delivery cycle by 50%.
- **Maintained automated PDK regression testing infrastructure** to ensure consistent release quality across multiple process nodes. Managed the transition from manual verification to automated standards, overseeing daily execution and troubleshooting.
- Collaborated frequently with the U.S.-based engineering leadership and cross-functional teams across North America and APAC; fluent in distributed engineering execution.

Senior Engineer — Device Modeling & EDA Enablement | UMC, Hsinchu, Taiwan *Jan 2018 – Mar 2020*

- **Delivered 10+ production-grade SPICE models** (MOSFETs, RF devices, varactors, diodes, resistors) deployed across UMC customer design flows for multiple technology nodes.
- Established BSIM4 extraction methodology documentation; built team mentorship program accelerating modeling team ramp-up and standardizing knowledge transfer practices.

EDUCATION

M.S. in Electronics | National Chiao Tung University, Hsinchu, Taiwan *Sep 2015 – Aug 2017*

First-author publications: IEEE Journal of Lightwave Technology; Proc. IEEE OMN 2018 — SPAD characterization in 0.18μm HV CMOS

B.S. in Physics | National Taiwan Normal University, Taipei, Taiwan *Sep 2011 – June 2015*

CERTIFICATIONS & LANGUAGES

Certifications: Cadence Advanced SKILL Language Programming · Cadence SKILL Development of Parameterized Cells

Languages: English (Fluent) · Mandarin (Native) · German (A2)